Geometry Honors Unit 5 Assessment Guide

General Information	Student learning of triangle congruence in this unit is supported by benchmarks MA.912.GR.2.1, MA.912.GR.2.3, MA.912.GR.2.6, and MA.912.GR.5.1. These benchmarks are not on the unit assessment because they provided the student with connections to their prior learning to support their understanding of MA.912.GR.1.2 and MA.912.GR.1.6. The benchmark MA.912.GR.1.2 is naturally supported by MA.912.LT.4.10 since the benchmark states that students are to prove triangle congruence. The draft test item specifications state that when assessing MA.912.GR.1.2 for triangle congruence, the student will only need to identify a reason that is stated in the benchmark. For this reason, the assessment does not have the student fully complete proofs. Teachers can edit the assessment so that students complete the proofs with all of the missing statements and reasons if that is preferred. Question 10 is assessing the honors benchmark by having students justify that using the given information is sufficient to show congruence by applying the definition of congruence in terms of rigid motions.
Calculator Information	Questions 1-7 on this assessment are calculator neutral so the assessment can be divided into a calculator and a non-calculator part, where questions 8 and 9 are on the calculator part of the assessment. Alternatively, teachers can allow students access to calculators for the entire assessment.
Total Recommended Points	The total recommended points in this guide are 100 points. Teachers can use the extra points if they choose to have students fully complete the proofs on this assessment.

Question	1	Benchmark(s)	MA.912.GR.1.2
Yes; ASA		**Recommended Scoring** **(6 Total Points)**	Consider giving partial credit (3 points) for students who correctly answer "yes" but give the incorrect theorem or postulate.
Question	**2**	**Benchmark(s)**	MA.912.GR.1.2
Yes; SSS		**Recommended Scoring** **(6 Total Points)**	Consider giving partial credit (3 points) for students who correctly answer "yes" but give the incorrect theorem or postulate.
Question	**3**	**Benchmark(s)**	MA.912.GR.1.2
Yes; AAS		**Recommended Scoring** **(6 Total Points)**	Consider giving partial credit (3 points) for students who answer "yes" but give the incorrect theorem or postulate.

Question	4	Benchmark(s)	MA.912.GR.1.2
It is given that $\overline{MC} \cong \overline{LC}$ and $\angle MCX \cong \angle XCL$. By the reflexive property, $\overline{XC} \cong \overline{XC}$. Therefore, $\Delta MCX \cong \Delta LCX$ by SAS.		**Recommended Scoring** **(11 Total Points)**	Consider giving partial credit for each step (3 points for $\overline{XC} \cong \overline{XC}$, 4 points for correctly naming the congruent triangles and 4 points for the correct congruence theorem.
Question	**5a**	**Benchmark(s)**	MA.912.GR.1.6
$\angle DFL \cong \angle DYL$.		**Recommended Scoring** **(5 Total Points)**	Partial credit is not recommended.
		Notes	This question is assessing student understanding of congruent parts of congruent triangles.
Question	**5b**	**Benchmark(s)**	MA.912.GR.1.2
$\overline{NW} \cong \overline{NR}$		**Recommended Scoring** **(5 Total Points)**	Partial credit is not recommended.
		Notes	This question is assessing student understanding of HL.

Question	**6**	**Benchmark(s)**	MA.912.GR.1.2
Reason for step 5	SAS	**Recommended Scoring** **(7 Total Points)**	Teacher may consider increasing the number of points if desiring students to complete all missing statements and reasons.
		Notes	This proof purposively applies student understanding of CPCTC to complete the proof.

Question	**7**	**Benchmark(s)**	MA.912.GR.1.2
Missing Reason	ASA	**Recommended Scoring** **(7 Total Points)**	Teacher may consider increasing the number of points if desiring students to complete all missing statements and reasons.
		Notes	Teachers may choose to have their students complete the entire proof.

<table>
<tr><th>Question</th><th>8a</th><th>Benchmark(s)</th><th>MA.912.GR.1.6</th></tr>
<tr><td colspan="2" rowspan="2">$x = 7.2$</td><td>Recommended Scoring

(8 Total Points)</td><td>Consider giving partial credit (4 points) for correctly setting $NW = DH$ or $RD = TB$ to determine the value of x. This shows an understanding of congruent parts. If a student correctly set up the equations but did not get the correct answer, consider their work. Use the remaining 4 points to distribute as partial credit.</td></tr>
<tr><td>Notes</td><td>Students who make a minor error in solving the equation is showing more content knowledge than a student who does not know how to solve the equation. Consider supporting students who show that they do not understand how to solve equations.</td></tr>
</table>

Question	8b	Benchmark(s)	MA.912.GR.1.6
<table><tr><th>Perimeter</th><td>70</td></tr></table>		**Recommended Scoring** **(10 Total Points)**	Students can make minor errors when determining the perimeter. Students who correctly show $TW + NW + NB + RY$ or $RD + DH + YH + RY$ is showing content knowledge of congruent figures. This should be weighed more than making an error in evaluating. It is suggested that students who make a minor error in evaluating earns 8 out of the 10 points.
Question	**9a**	**Benchmark(s)**	MA.912.GR.1.6
$m\angle TLC = 66°$		**Recommended Scoring** **(5 Total Points)**	Students should match $\angle TLC$ to $\angle GXR$.

Question	9b	Benchmark(s)	MA.912.GR.1.6
$m\angle KGX = 120°$		**Recommended Scoring** **(10 Total Points)**	Student work is important to monitor. It is in their work that the teacher can determine if the student correctly connected the congruent angles. Students who show $2(3x+18)+2(3x-8)+90+66=720$ have correctly determined all of the corresponding congruent angles. Any errors that follow are algebraic in nature. Although knowing how to solve the equation is important, it is suggested that students earn most of their points in the correct setup. It is suggested that students who make a minor error in solving earns 8 out of the 10 points.

Question	10a	Benchmark(s)	MA.912.GR.2.6
Translate ΔRBH to the left 2 units and down 1 unit, then rotate $90°$ clockwise about point R		**Recommended Scoring** **(4 Total Points)**	Students may state transformations using different methods. Consider partial credit for minor errors.
Question	**10b**	**Benchmark(s)**	MA.912.GR.2.7
Complete the statements. Because the mapping described consists of [o dilations, o rigid motions, o both dilations and rigid motions,] the definition of [o similarity o congruence] can be applied to show that [o the vertices o corresponding sides o corresponding angles o both corresponding sides and corresponding angles] are [o similar. o congruent.] Therefore, since $\overline{HB} \cong \overline{WM}$, $\angle WMT \cong \angle HBR$, and $\angle HRB \cong \angle WTM$ were given, then [o AAS o ASA o SAS o SSS] is sufficient to show triangles are [o similar. o congruent.]		**Recommended Scoring** **(10 Total Points)**	Consider distributing the points as 5 points for each correctly completed statement. It is not suggested that students get partial credit for single correct answer boxes within each statement.